

# Göksenin Yüksel

PhD Candidate in AI · Machine hearing, representation learning and spatial audio

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## Research Interests

**Machine hearing:** Models that understand real-world sound scenes: what is heard and where it comes from

**Representation learning:** Self-supervised learning (masked autoencoders, JEPA), audio foundation models, robustness to noise and reverberation

**Spatial audio:** Ambisonics, binaural audio, geometry-agnostic spatial representations for arbitrary microphone arrays, sound event localization and detection (SELD), equivariant models, spatial audio large language models

**Speech:** Compute-efficient self-supervised speech representation learning, automatic speech recognition (ASR)

## Education

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| <b>PhD</b> | <b>Donders Institute, Radboud University</b> , Artificial Intelligence, Machine Hearing<br>• Research: machine hearing, self-supervised (spatial) audio and speech representation learning, and spatial audio LLMs.<br>• AudioSphere (NeurIPS 2026): first sound event localization and detection from a frozen self-supervised backbone; 1.9° median direction-of-arrival error on a planar microphone array.<br>• SpeechJEPA beats wav2vec 2.0 with one hour of labels while using about 17× fewer training FLOPs.<br>• Bearings raises the location-aware SELD F-score from below 4 to 50 on TAU-NIGENS 2021.<br>• Supervisors: Kiki van der Heijden and Marcel van Gerven. Member of the ELLIS Unit Nijmegen. | Nijmegen, NL<br>Sept 2024 – Jan 2028   |
| <b>MSc</b> | <b>University of Amsterdam</b> , Artificial Intelligence<br>• GPA: 4.0/4.0.<br>• Thesis: Domain Adaptation of Dense Retriever Models.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | Amsterdam, NL<br>Sept 2022 – Sept 2024 |
| <b>BSc</b> | <b>Radboud University</b> , Artificial Intelligence<br>• GPA: 3.96/4.0.<br>• Teaching assistant for Functional Programming and Bayesian Statistics.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | Nijmegen, NL<br>Sept 2019 – Sept 2022  |

## Grants

**Snellius compute grants:** Three grants (EINF-17264, EINF-14624, EINF-12218) totaling 3 million SBUs, about 15,000 H100 GPU-hours on the Dutch national supercomputer (SURF)

## Peer-Reviewed Publications

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| <b>AudioSphere: Towards Self-Supervised Spatial Audio Representation Models</b><br>Göksenin Yüksel, Marcel van Gerven, Kiki van der Heijden<br><a href="https://github.com/labhamlet/AudioSphere">github.com/labhamlet/AudioSphere</a> (NeurIPS 2026)                                                                                                  | Sept 2026 |
| <b>University of Amsterdam at the CLEF 2024 SimpleText Track</b><br>Jan Bakker, Göksenin Yüksel, Jaap Kamps<br><a href="https://ceur-ws.org/Vol-3740/paper-310.pdf">ceur-ws.org/Vol-3740/paper-310.pdf</a> (Working Notes of CLEF 2024, CEUR Workshop Proceedings 3740)                                                                                | Sept 2024 |
| <b>Style Randomization Improves the Robustness of Breast Density Estimation in MR Images</b><br>Göksenin Yüksel, Koen Eppenhof, Jaap Kroes, Marcel Worring<br><a href="https://proceedings.mlr.press/v250/yuksel24a.html">proceedings.mlr.press/v250/yuksel24a.html</a> (MIDL 2024 (PMLR 250))                                                         | July 2024 |
| <b>Reproducibility Study of “Quantifying Societal Bias Amplification in Image Captioning”</b><br>Farrukh Barátov*, Göksenin Yüksel*, Darie Petcu*, Jan Bakker*<br><a href="https://openreview.net/forum?id=eJmQJT0Dtt">openreview.net/forum?id=eJmQJT0Dtt</a> (NeurIPS 2023 Main Hall Poster, ML Reproducibility Challenge 2022; * equal contribution) | Dec 2023  |

## Under Review

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| <b>SpeechJEPA: Clustering-Free Latent Prediction for Compute-Efficient Speech Representation Learning</b><br><b>Göksenin Yüksel</b> , Marcel van Gerven, Kiki van der Heijden<br><a href="https://github.com/labhamlet/SpeechJEPA">github.com/labhamlet/SpeechJEPA</a> (Under review at ICASSP 2027)             | Sept 2026 |
| <b>Bearings: Self-Supervised Soundfield Embeddings from First-Order Ambisonics</b><br><b>Göksenin Yüksel</b> , Marcel van Gerven, Kiki van der Heijden<br><a href="https://arxiv.org/abs/2609.23152">arxiv.org/abs/2609.23152</a> (Under review at ICASSP 2027)                                                  | Sept 2026 |
| <b>EquiSELD: Efficient Training of Equivariant Sound Event Localization and Detection Networks</b><br><b>Göksenin Yüksel</b> , Marcel van Gerven, Kiki van der Heijden<br><a href="https://arxiv.org/abs/2609.23156">arxiv.org/abs/2609.23156</a> (Under review at ICASSP 2027)                                  | Sept 2026 |
| <b>WavJEPA: Semantic Learning Unlocks Robust Audio Foundation Models for Raw Waveforms</b><br><b>Göksenin Yüksel</b> , Pierre Guetschel, Michael Tangermann, Marcel van Gerven, Kiki van der Heijden<br><a href="https://arxiv.org/abs/2509.23238">arxiv.org/abs/2509.23238</a> (Under review at IEEE/ACM TASLP) | Sept 2025 |

## Preprints

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| <b>GRAM: Spatial General-Purpose Audio Representations</b><br><b>Göksenin Yüksel</b> , Marcel van Gerven, Kiki van der Heijden<br><a href="https://arxiv.org/abs/2506.00934">arxiv.org/abs/2506.00934</a>   | June 2025 |
| <b>Remining Hard Negatives for Generative Pseudo Labeled Domain Adaptation</b><br><b>Göksenin Yüksel</b> , David Rau, Jaap Kamps<br><a href="https://arxiv.org/abs/2501.14434">arxiv.org/abs/2501.14434</a> | Jan 2025  |
| <b>Interpretability Analysis of Domain Adapted Dense Retrievers</b><br><b>Göksenin Yüksel</b> , Jaap Kamps<br><a href="https://arxiv.org/abs/2501.14459">arxiv.org/abs/2501.14459</a>                       | Jan 2025  |
| <b>On Correlating Factors for Domain Adaptation Performance</b><br><b>Göksenin Yüksel</b> , Jaap Kamps<br><a href="https://arxiv.org/abs/2501.14466">arxiv.org/abs/2501.14466</a>                           | Jan 2025  |

## Experience

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| <b>ScreenPoint Medical</b> , AI Research Intern<br>• Built a robust 3D MRI segmentation pipeline for breast density estimation, now used for internal R&D.<br>• Proposed a style-randomization augmentation for domain generalization across MRI sequences; first-author paper at MIDL 2024.           | Nijmegen, NL<br>Sept 2023 – Feb 2024<br>6 months        |
| <b>ScreenPoint Medical</b> , Software Engineer (part-time)<br>• Developed and maintained automated tests for large-scale medical AI software (2D/3D medical image processing).<br>• Automated manual verification steps and led a project to produce a shorter, more maintainable verification report. | Nijmegen, NL<br>Jan 2023 – Sept 2024<br>1 year 9 months |
| <b>Arcadis</b> , Data Science Intern<br>• Built and deployed machine learning dashboards for housing price analysis and for the optimal placement of EV parking spots and charging poles.<br>• Presented findings to stakeholders and advised on data-driven business decisions (Python, SQL).         | Amsterdam, NL<br>July 2021 – Oct 2021<br>4 months       |

## Technical Skills

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| <b>Deep learning:</b> PyTorch, PyTorch Lightning, Transformers, FlashAttention, large language models (LLMs)               |
| <b>Distributed training:</b> Large-scale GPU cluster training; efficient multi-node scaling with FSDP and DDP via torchrun |
| <b>Audio &amp; signal processing:</b> Ambisonics, binaural audio, microphone arrays, room acoustics simulation             |
| <b>Languages &amp; tools:</b> Python, Java, Bash, Linux, Docker, Git, Apache Spark; AI coding agents (Claude Code, Codex)  |